Multi-Factor ETF Strategy

Operating Document

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Abstract

Long-only smart-beta ETF strategy operated against a live IB Gateway account. Selects positions from a filtered universe of ∼5,000 US-listed ETFs by ranking on a weighted geometric mean of four factors (momentum, quality, low-volatility, value). Top 30 held with exponential rank weights; 10% trailing stops on every position; drift-triggered rebalance (∼6/yr). This document is the operator's reading surface and takes precedence over marketing summaries; the code in `src/` is authoritative for behaviour, this document is authoritative for intent.

Contents

1 Executive Summary

1.1 What This Is

A systematic, deterministic, long-only ETF portfolio. No LLM sets sizes or places orders. No factor timing. No sentiment trading. All numeric parameters trace to either academic literature or the calibration diagnostic under `docs/research/`.

Operated weekly: cache refresh Saturday evening, review Sunday, apply mid-week when the model proposes a rebalance and the operator approves.

1.2 Live Snapshot (2026-07-09)

Net Liquidation Value	\$390,940
Cash	\$84,552
Positions	28
Live earnings (owner's estimate)	∼13–14%

Position count of 28 sits below the new target of 30 (raised from 20 on 2026-07-10). The next rebalance will add positions to close the gap and redeploy the freed \$40k from the reduced cash reserve.

1.3 Backtest Metrics (35/30/20/15 policy, top-20, RankBased exponential)

Level-1 diagnostic `docs/research/2026-07_factor_weight_diagnostic.md` on the cached ETF universe 2021-04-12 to 2026-04-16, monthly rebalance, 5 bps turnover.

Window	Sharpe	Sortino	CAGR	Max Drawdown
Full 5-year	0.47	0.60	8.07%	-9.55%
Trailing 3-year	0.96	1.22	13.32%	-9.09%

The live earnings estimate matches the 3-year backtest CAGR. The 5-year window includes the 2022 bear market drawdown; the 3-year excludes it.

1.4 Positioning

The strategy targets modest outperformance versus SPY with meaningfully lower drawdown. It is not designed to beat SPY by 5% per year. A year in which the strategy returns 12% while SPY returns 15% is a good year for the strategy *if* its drawdown is materially lower over the same period.

2 Strategy Mechanics

2.1 Universe Selection

Starting from the full IB-listed ETF universe (˜5,000 tickers), the filter chain is:

1. History requirement: ≥252 trading days of price data.
2. Data quality: <10% missing daily bars.

3. Leveraged/inverse exclusion: leveraged ETFs are removed by ticker prefix and issuer classification.
4. Result: ∼2,000 candidates on which factor scores are computed.

2.2 Factor Framework — What the Four Factors Are and Why

The four factors below are the AQR-canonical smart-beta four: momentum, quality, low-volatility, value. They were chosen because they are the four with the largest and most robust empirical premia over the widest range of markets and time periods studied in the academic literature. AQR chose these because after decades of empirical work they are the return sources that survive out-of-sample testing, transaction costs, and factor-crowding. The strategy does not attempt to identify novel factors.

2.2.1 Momentum (target 35% weight)

$$\text{Momentum}_i = \frac{P_{i,t-21} - P_{i,t-252}}{P_{i,t-252}} \tag{1}$$

12-month return, with the most recent 21 days omitted. Winsorised at the 1st and 99th percentile.

Why: Assets that have outperformed over the prior year tend to continue outperforming over the next year. The 1-month skip avoids the short-term reversal effect that would contaminate the trend signal. Established by Jegadeesh & Titman (1993); the skip refinement from Novy-Marx (2012).

Why AQR uses it: The most-studied, most-robust factor after value, and the largest-premium factor in the last decade. Works across equity, futures, and FX. Fails during trend reversals (2022 tech drawdown, 2000 dot-com peak).

2.2.2 Quality (target 30% weight)

Composite of three z-score-normalised components on 252-day windows:

$$\text{Sharpe Ratio} = \frac{\mu_r - r_f}{\sigma_r}\sqrt{252} \quad \text{(40\% weight)} \tag{2}$$

$$\text{Drawdown Resilience} = -1 \times \text{MaxDD} \quad \text{(30\% weight)} \tag{3}$$

$$\text{Return Stability} = -1 \times \sigma_r\sqrt{252} \quad \text{(30\% weight)} \tag{4}$$

Why: Assets that produce stable, high risk-adjusted returns tend to continue doing so. Established by Asness, Frazzini & Pedersen (2019).

Why AQR uses it: Quality captures the defensive-equity premium. Correlated with low-volatility but not identical — quality picks up profitability and stability of *returns*, low-vol picks up stability of *prices*. Both belong in the portfolio because they hedge different aspects of stress.

2.2.3 Low-Volatility (target 20% weight)

$$\text{Volatility}_i = \frac{1}{\sigma_{60d,i} \times \sqrt{252}} \tag{5}$$

Inverse of annualised realised volatility over 60 trading days. Higher score for lower-volatility ETFs.

Why: The low-volatility anomaly — lower-volatility assets have historically produced higher risk-adjusted returns than higher-volatility assets, contradicting the CAPM. Baker, Bradley & Wurgler (2011).

Why AQR uses it: Two reasons. First, the low-vol premium is real and persistent. Second, low-vol allocations reduce portfolio-level whipsaw, which matters more than arithmetic-mean metrics suggest in a compounding portfolio (see §3).

2.2.4 Value (target 15% weight)

$$\text{Value}_i = 0.60 \cdot z(\text{DividendYield}_i) + 0.40 \cdot z(-\text{ExpenseRatio}_i) \tag{6}$$

What each component contributes. Dividend yield is the canonical fund-level value tilt for ETFs — funds like SCHD, VYM, HDV, and VTV that hold value-tilted holdings publish materially higher trailing yields than growth-oriented funds like VUG, VGT, and QQQ. It is a real cross-sectional signal, not a fee preference. Expense ratio contributes the cost-efficiency component: for two funds with similar yield, the cheaper one is a better long-run compounder. Both components are z-score-normalised on the current cross-section before the blend.

Data source. Real per-fund dividend yield is computed as `lastDividend/price` from Financial Modeling Prep's `stable/profile` endpoint. Real per-fund expense ratio comes from the `stable/etf/info` endpoint. Both are cached weekly to `~/trade_data/ETFTrader/processed/etf_fundamenta` by `scripts/refresh_fundamentals.py`. Rankings verified on 2026-07-10: SCHD 1.04, HDV 0.81, VYM 0.56, VTV 0.38, VOO −0.04, VUG −0.39, VGT −0.50, ARKK −1.81 — the expected value ordering.

Residual gap. Fund-level weighted P/E and P/B are still absent. The FMP endpoints that publish these (`key-metrics-ttm`, `ratios-ttm`) aggregate 10-K filings and return zero rows for ETFs at any subscription tier because ETFs do not file 10-Ks. Adding true weighted-average P/E would require holdings-level aggregation via `etf/holdings` (Ultimate tier) or a Morningstar-class source. Tracked as T1.1 residual in §4. Current implementation delivers approximately 60% of the intended value composite.

Why AQR uses value: It is the oldest and most-studied factor (Fama & French 1993). Tends to work when growth and momentum break (2000–2002, 2022). Yield-based value is a defensible, slowly-varying subset of the full academic tilt.

2.3 Factor Integration

$$\text{Score}_i = \text{Mom}_i^{0.35} \times \text{Qual}_i^{0.30} \times \text{Vol}_i^{0.20} \times \text{Val}_i^{0.15} \tag{7}$$

Each factor is first converted to a percentile rank in [0,1]. The integrated score is the weighted geometric mean of ranks.

Why geometric mean, not arithmetic: Geometric-mean integration penalises weakness on any one factor. An ETF in the 90th percentile on three factors but the 20th on one ranks below an ETF in the 70th percentile on all four. This "multi-factor consistency" requirement — explicitly the key insight from AQR's factor-integration research — rejects one-trick candidates in favour of broadly-strong candidates. Arithmetic integration would allow one strong factor to compensate for weakness on the others.

2.4 Portfolio Construction

1. Score every ETF in the filtered universe.
2. Rank descending.
3. Take the top 30 (raised from 20 on 2026-07-10 — see Change Log).
4. Assign exponential rank weights: $w_i \propto e^{-\text{rank}_i/30}$, normalised to sum to 1.
5. Apply weight bounds: minimum 2%, maximum 15%.

Exponential weighting expresses conviction — the top-ranked position gets meaningfully more weight than the thirtieth — while the bounds prevent single-position concentration. The alternative optimiser (`OPTIMIZER_TYPE = "mvo"`) is available in the code but not the live default.

2.5 Rebalancing

1. Scheduled: bimonthly (~6 times per year).
2. Drift-triggered: if any position's actual weight deviates from target by >5%, rebalance.
3. In practice, drift triggers dominate; scheduled rebalances rarely fire independently.

Drift threshold is deliberately wide. Tighter thresholds cause more trading on smaller deviations — textbook whipsaw. See §2.8.

2.6 Risk Management

Every entry BUY places a 10% trailing stop, GTC, on the position. Trailing stops ratchet up as the position appreciates. On any 10% pullback from the running high, the stop triggers.

Entry stop-loss of 12% (a bit wider than the trailing stop) accommodates initial-position volatility before the trailing high-water mark establishes.

Cash reserve of $30,000 is held unallocated (lowered from $70,000 on 2026-07-10 to free capital for the widened 30-position basket). This is a static allocation, not a regime-responsive hedge; the T2.1 regime overlay in progress will replace the static reserve with a dynamic risk-off multiplier.

2.7 Explicit Assumptions

The strategy assumes:

- Factor premia (momentum, quality, low-vol, value) will continue to exist over the holding horizon. They may attenuate but do not disappear.
- The rank structure of the universe changes slowly enough that monthly rebalancing captures the signal without significant slippage.
- Retail transaction costs at IB (~5 bps effective) are small enough not to erode the premium.
- Long-only holds; no short exposure to hedge market drawdown.
- No leverage.
- Trailing stops execute at or near their trigger prices. Gap-down scenarios (weekend news, halt reopens) are unhedged.
- Position selection ignores tax lot-level implications. Tax-lot management is out of scope.

2.8 Design Principles

Three non-negotiable filters govern every future proposal:

1. **Deterministic decides.** Every number reaching the broker comes from code in `src/`. No LLM output influences a size or weight.
2. **No magical thinking on knobs.** Every parameter traces to research or a repo-recorded backtest. Knobs whose provenance is intuition go on the research backlog.
3. **Slowly varying only.** Whipsaw is evidence of overfitting, not alpha. Long lookbacks are preferred over short. Drift thresholds err high. Regime signals (when added) require hysteresis and minimum dwell time. Factor timing is rejected on principle.

3 Parameters and Weights

3.1 The Factor Weights

The weight vector is momentum = 35%, quality = 30%, low-volatility = 20%, value = 15%. This is an *informed prior*, deliberately regularised away from any in-sample optimum.

3.1.1 Why not the in-sample winner?

The July 2026 diagnostic (grid search over the three-factor simplex, momentum/quality/low-vol, on the cached 2021–2026 history) produced in-sample winners uniformly momentum-heavy: 80–90% momentum, 0–30% quality, 10–20% low-vol. Sharpe up to 1.30 on the 3-year window versus $\approx$0.96 for the current policy.

This finding was deliberately not acted on. Three reasons:

1. **Single-regime evidence.** 2021–2026 was overwhelmingly momentum-friendly (post-COVID rally, mega-cap tech, AI narrative). Both the 5-year and 3-year windows overlap and sample the same regime. This is not an independent invariance test.
2. **Non-ergodicity.** Equity returns are multiplicative. For a single investor on a single path, the arithmetic mean of returns diverges from the time-average growth rate they actually experience (Peters 2019). A 50% drawdown requires 100% subsequent gain to recover, not 50%. In-sample-Sharpe optimisation understates this asymmetry. Factors that reduce drawdown variance (quality, low-vol) buy protection against the multiplicative losses a single-path investor most needs to avoid. These are insurance premia, not underperforming factors.
3. **Confirmation bias.** The operator is explicitly on record as most concerned about an AI/tech valuation regime unwind. The mechanical winner (90% momentum) is exactly the exposure that would be worst affected in that scenario. Adopting the in-sample winner would be confirming a bias against the operator's stated risk view.

3.1.2 The Constraint Framework

The 35/30/20/15 vector satisfies three constraints:

1. **Ordering: momentum > quality > low-vol > value.** Reflects the ordering of the empirical premia in the AQR literature and the compounding effect that the value implementation is currently a low-fee proxy rather than a true value factor.
2. **No factor below 10%.** A factor below 10% receives such a small weight in the weighted geometric mean that it fails to discriminate between candidates. The multi-factor-consistency requirement collapses to a three-factor screen.
3. **No factor above 40%.** A factor above 40% dominates the integrated score to the point that single-factor selection would produce nearly the same portfolio. This defeats the multi-factor rationale and re-introduces single-factor tail risk.

Within those constraints the exact values are round numbers chosen for auditability. They can be reproduced in one sentence to a future maintainer: "momentum gets the largest weight but capped, quality gets the second-largest, low-vol gets a solid third to hedge whipsaw, value gets the smallest until we fix the proxy."

3.1.3 When To Revise

Revise the weights when at least one of the following is available:

- Backfilled history to pre-2020 that permits a non-overlapping regime comparison (2011–2015 vs 2021–2026).
- A fund-level P/E or P/B signal added to the value composite (yield + expense ratio blend committed 2026-07-10 partially satisfies this; full P/E/P/B would fully satisfy).
- A deliberate operator-declared factor-level sentiment override, committed to `configs/etf_smart_beta.tc` with a dated research note.

3.2 Portfolio Construction Parameters

Parameter	Value	Justification
`num_positions`	30	Target holding count. Prior operating configurations without a fixed target produced degenerate 1–2 position portfolios when factor scores were bimodal; the fixed target forces diversification. Raised from 20 to 30 on 2026-07-10 to spread capital more thinly across the top-ranked cohort and reduce single-name idiosyncratic risk in a sideways market.
`min_weight`	2%	Position floor. At 30 positions the full basket uses 60% of NAV at floor, leaving 40% for over-weight tilts on high-conviction positions.
`max_weight`	15%	Position cap. Prevents single-position concentration under the RankBased scheme; the top-ranked ETF cannot receive more than 15% of NAV.
`tax_status`	`taxable`	Signals real capital-gains consequences to any future tax-lot / wash-sale logic.

3.3 Factor Calculation Windows

Parameter	Value	Justification
Momentum lookback	252 days	Jegadeesh & Titman (1993). 12-month formation window.
Momentum skip	21 days	Novy-Marx (2012). Avoid short-term reversal.
Quality lookback	252 days	AQR convention.
Value lookback	252 days	Unused while value is expense-ratio proxy.
Volatility lookback	60 days	~3 months of daily returns. Standard practice.

3.4 Rebalancing Parameters

Parameter	Value	Justification
Scheduled frequency	bimonthly (~6/yr)	Balances signal freshness against turnover.
Drift threshold	5%	Round number, defensible per Donohue–Yip (2003). Widening (7.5%, 10%) considered before tightening.

3.5 Risk Management Parameters

Parameter	Value	Justification
Entry stop-loss	12%	Slightly wider than trailing stop to accommodate initial position volatility.
Trailing stop	10%	Sized to sit above ordinary noise (≈2–3σ daily) so it triggers only on genuine trend breaks.
Cash reserve	$30,000	Lowered from $70k to $30k on 2026-07-10. Roughly 8% of current NAV; freed $40k for redeployment into the 30-position basket. Static, not regime-responsive — fix on the list (§4 T2.1).

3.6 Optimiser Priors (MVO variant only)

The live default optimiser is RankBased. The values below apply only when `OPTIMIZER_TYPE = "mvo"`.

Parameter	Value	Justification
Risk aversion λ	1.5	Standard MVO. Meucci retail range 2–4; 1.5 is risk-tolerant end.
Robustness penalty γ	0.7	Axioma-style, aggressive. Chosen default.
Turnover penalty	0.2	Medium. Chosen default.

4 Current Flaws

Adversarial review, tiered by impact. Items remain here until closed; closed items move to the Change Log with the closing commit.

4.1 Tier 1 — material design flaws

- **T1.1 The value factor is fake. Partially closed 2026-07-10.** `ValueFactor` now consumes real per-fund dividend yield (from FMP `stable/profile`, `lastDividend/price`) and real per-fund expense ratio (from FMP `stable/etf/info`). Blend is 60% yield + 40% expense ratio. Rankings verified sensible: SCHD/HDV/VYM/VTV score at the top, VUG/VGT/ARKK at the bottom. **Residual gap:** fund-level weighted P/E and P/B remain unavailable at any FMP tier (`key-metrics-ttm` and `ratios-ttm` return zero rows for ETFs since they aggregate 10-K filings; `etf/holdings` needs Ultimate tier and thousands of aggregation calls). A future upgrade path exists via Ultimate-tier holdings aggregation or a Morningstar-class source. Current implementation delivers roughly 60% of the intended value composite — the yield signal alone is the canonical academic value tilt for ETFs.

- **T1.2 Survivorship bias in the cached universe — accepted as small.** The per-ticker parquet cache contains only ETFs that still trade. Delisted / liquidated ETFs are absent. In academic stock-market backtests survivorship bias inflates CAGR by 1–2%, but for this specific strategy the effect is materially smaller: the momentum (252d) and quality (252d) screens would have filtered out most now-delisted ETFs long before they liquidated, and the top-30 selection would not have held them. Operator judgement (2026-07-10): the residual bias is not worth the engineering cost to correct while the model is untested. Revisit only if a live regime materially favours niche or newly-launched ETFs.

- **T1.3 Look-ahead bias presumed absent, never proved.** No pinning test asserts that factor calculations at time t use only data with index $< t$. If any factor uses a window that includes $t{+}1$, the backtest is silently inflated. Fix: add a shift-and-recompute test. Two hours' work.

4.2 Tier 2 — significant weaknesses

- **T2.1 No regime awareness. In progress 2026-07-10.** Strategy runs identically in bull, bear, and crisis regimes. Long-only, concentrated, static ≈8% cash reserve ($30k on $390k NAV), 10% trailing stops. Signal to de-risk in rising-VIX / breadth-collapsing environments is being built as `src/portfolio/regime.py` (committed, unit-tested). VIX history 2010–2026 cached via FMP `historical-price-eod/full`. Signal: SPY > 200-day SMA AND VIX < 25, smoothed by a 10-day hysteresis window, minimum 30-day dwell between switches. When risk-off, target equity multiplied by 0.60 (cash rises to ≈40% of NAV). Backtest across the 2010–2026 regime record is the next step before the overlay is enabled live.

- **T2.2 Correlation blindness in top-N selection.** The integrated score picks the top 30 ETFs by factor score without checking their pairwise correlations. Current live holdings read as a broad international dividend / value / quality tilt with likely 0.6–0.8 pairwise correlations. Effective diversification is materially less than 30 positions suggests. Fix: take top 60 by score, cluster on 6-month correlations, pick top-scored representative per cluster until 30 positions filled. FMP `etf/info` sector data (newly available on Premium) enables a sector-aware variant of the clustering. Design note at `docs/research/2026-07_correlation_clustering_design.md`.

- **T2.3 Conviction magnitude discarded.** The integrated score's magnitude (0.95 vs 0.62) is not used in the optimiser; only rank order matters. Exponential rank weighting mitigates partially. Fix: pass score magnitude as expected-return alpha to the optimiser.

4.3 Tier 3 — smaller nudges

- **T3.1 Factor lookback windows are convention.** 252 / 21 / 60 chosen by AQR convention, never tested for optimality in this universe.

- **T3.2 Risk parameters are round numbers.** 10%, 12%, $30k, 5% drift, 2–15% weight bounds — all defensible, all uncalibrated. Sensitivity grid would locate them on the risk-return frontier.

- **T3.3 Rebalance frequency untested.** Bimonthly vs quarterly turnover trade-off never explicitly measured on this universe.

4.4 Priority Order

1. T2.1 (regime overlay) — **in progress**. Module and unit tests committed 2026-07-10; backtest across 2010–2026 pending. Drawdown protection before live orders resume.
2. T2.2 (correlation clustering) — **next**. Effective diversification; will use FMP sector data for a sector-aware variant.

3. T1.3 (look-ahead pinning test) — 2 hours. Proves the model is not silently a lie. Do before enabling any of the above live.
4. T1.1 residual (fund-level P/E / P/B) — deferred. Requires FMP Ultimate tier holdings aggregation or a Morningstar-class source. Review after 3–6 months of live experience with the yield + expense-ratio blend.
5. T1.2 (survivorship haircut) — 1 hour. Honesty in reporting.
6. T2.3, T3. — as time permits.

5 Change Log

Chronological, most recent first. Each entry cites the closing commit SHA(s). Every commit touching `configs/etf_smart_beta.toml`, `src/factors/`, `src/portfolio/`, or the execution scripts must append an entry.

2026-07-10 — T1.1 partial close: real value factor via FMP Premium

Value factor rewired end-to-end. `src/factors/value_factor.py` now blends real per-fund dividend yield (60% weight) with expense ratio (40% weight). Fundamentals cache at `~/trade_data/ETFTrader/pr` refreshed weekly by `scripts/refresh_fundamentals.py` over the union of the defined universe and the price cache (~5000 tickers). Data source is Financial Modeling Prep Premium tier (~$44/mo; verified capabilities on 2026-07-10 — see rule_verify_before_recommending memo). Sanity check: SCHD scores 1.04, VYM 0.56, VTI −0.05, VUG −0.39, ARKK −1.81 — the expected value ordering. Residual: fund-level P/E and P/B remain unavailable at current FMP tier, documented in T1.1. Commit: 2d4c671 (endpoint verification), plus follow-up wiring.

2026-07-10 — Position count 20→30, cash reserve $70k→$30k

Operator directive following observation that the current 20-name basket has been drifting sideways with the market: raise `num_positions` to 30 and lower `cash_reserve` to $30,000 in `configs/etf_smart_beta.toml`. Rationale: spread capital more thinly across the ranked top-cohort to reduce single-name idiosyncratic drag, and free approximately $40k of dormant cash for redeployment. Passes the slowly-varying filter (parameter change, not signal change) and preserves the design invariants ($30 \times 2\%$ floor = $60\% < 100\%$; ranking and scoring logic untouched). Related open item: T2.2 correlation clustering — with a wider basket the marginal decorrelation benefit from clustering is expected to fall, and the experiment should quantify this.

2026-07-10 — Full rewrite

Document rewritten from the ground up. Historical version-comparison content (v3 → v4 migration story, v4.2 backtest tables, Feb-2026 holdings snapshot) removed. New structure: Executive Summary → Strategy Mechanics → Parameters and Weights → Current Flaws → Change Log. Factor definitions and integration rationale promoted into the Mechanics section; the constraint framework (10%–40% weight bounds, factor ordering) promoted into the Parameters section. Length approximately halved.

2026-07-09 — Adversarial review + weight justification (superseded by rewrite)

Adversarial review commissioned and recorded. Ergodicity-economics justification for retaining the 35/30/20/15 weight vector recorded in the Factor Framework section. Design Principles (deterministic decides, no magical thinking, slowly varying) recorded as non-negotiable filters. Level-1 factor-weight diagnostic produced at `docs/research/2026-07_factor_weight_diagnostic.md`. Commits d9826dc, 2198273, 6a14212.

2026-02-15 — Prior v4.2 design (historical, retained for context)

Prior design documented 30 positions, 3–8% weight bounds, quarterly rebalance, Robust MVO default. Backtested 12.1% CAGR / 0.64 Sharpe over 2021–2026. Live configuration diverged from this design in three places (position count, weight bounds, default optimiser) and the 2026-07 rewrite reconciled the document to what is actually running.